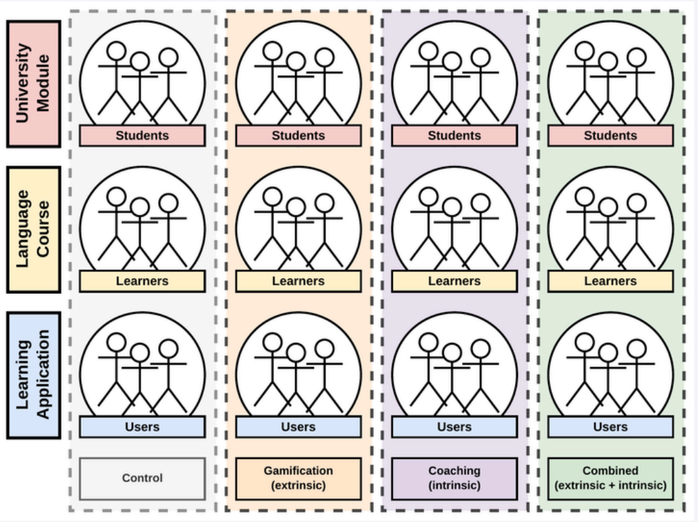
Outlining an Experiment Design of Gamifying Language Learning in Low-Resource Contexts

Kurdish Learning in University, Language Courses, and on Web-Based Applications

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3 learning environments × 4 condition families

Why Kurdish?

Only a very small number of languages are represented in language technologies and applications (Joshi et al., 2020).
Kurdish, in contrast, is considered a low-resource language

TECHNOLOGY

Language technology
few corpora and tools

Digital learning
largely unavailable for automated assessment or generated feedback

Abdulrahman et al. 2019; Ahmadi 2020; Ahmadi et al. 2024

TEACHING

Didactic materials
few high-quality materials available

Teaching research
little research on Kurdish teaching

Zhang 2026; Muhammadamin 2025; Resul and Muhammadamin, 2015

SPEAKERS

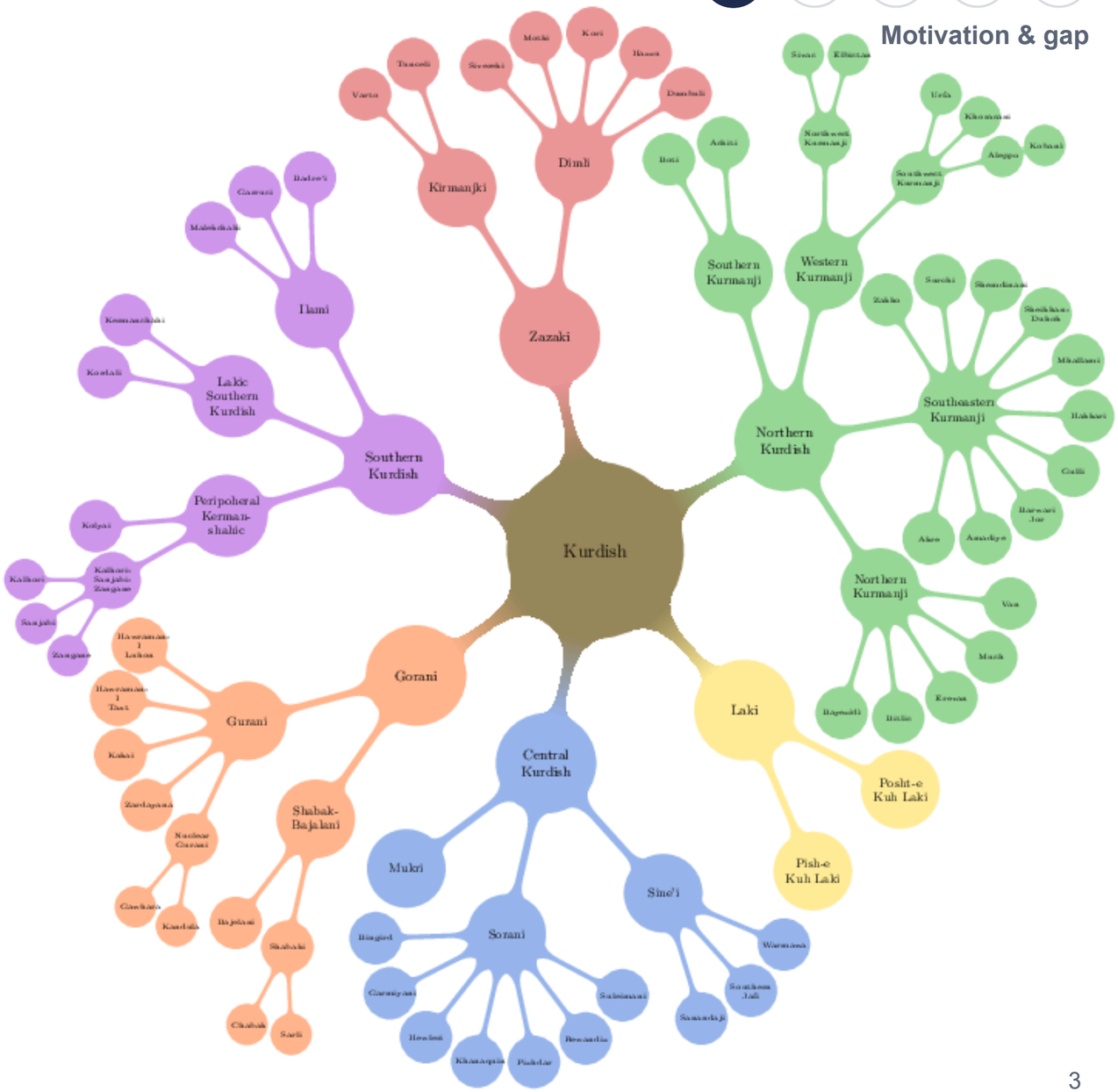
Limited use
few opportunities in everyday life

Generational attrition
dwindling incentives for younger speakers

Kurdish languages

Northern Kurdish (kmr)	
Standard	Ez westiyayî me
Bahdînî	Ez yê westiyayî me
Efrînî	E ri'eti me
Kobanî	Ez-î westiyayî me
Qamişlokî	Ez te'bî me
Khorasani	Ez westî me
Khorasani	Ez westiya me
Central Kurdish (ckb)	
Sulaymaniyah	ماندووم
Hawrami (hac)	
Hawraman	Manîyane
English	I am tired

Central Kurdish (ckb)	
Standard	من شوو ناکەم
Ardalanî	من ژن ناخوازم
Babanî	من ژن ناهێنم
Hewlêrî	ئەمن ژنی ناینم
Mukrî	ئەمن ژن ناهێنم
English	I don't marry



Some dialectal variations in Kurdish.
(Schuler et al., 2026)

Research gap

Gamification effects are inconsistent (Dichev and Dicheva, 2017; Sailer and Homner, 2020; Luo, 2023), and evidence comes mostly from well-resourced languages and learners with a clear instrumental motive.

LEARNER PERCEPTION

Little research

on how learners perceive learning activities and gameful design in low-resource language-learning contexts

LEARNER MOTIVES

Other motives

e.g., Chinese students: Belt and Road, cultural & literary ties, “beganeperestî”

+ intrinsic motivation

via a self-regulation agent

Chaziza 2024; World Bank 2019; Dadparvar & Shams 2024; Jasim 2024

TRANSFER

Across environments?

unclear: a student facing exams vs. a free-app user with very different costs of quitting

Vary the environment

keep the interventions comparable

Research questions



RQ1

Gameful redesign

How does gameful redesign of existing learning activities affect learning outcomes and self-reported motivation?



RQ2

Coaching agent

Does adding an agent that targets self-regulation and learning behavior yield effects beyond those of gameful design alone?



RQ3

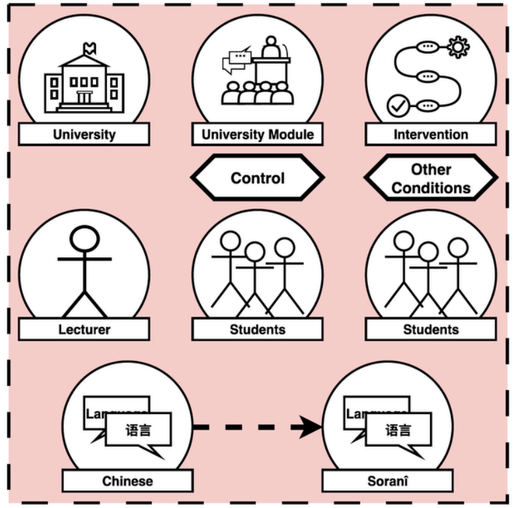
Learning environment

How do these effects differ across environments that vary in institutional pressure, financial investment, and voluntariness?

Experiment environments

Methodology: 3 × 4 design

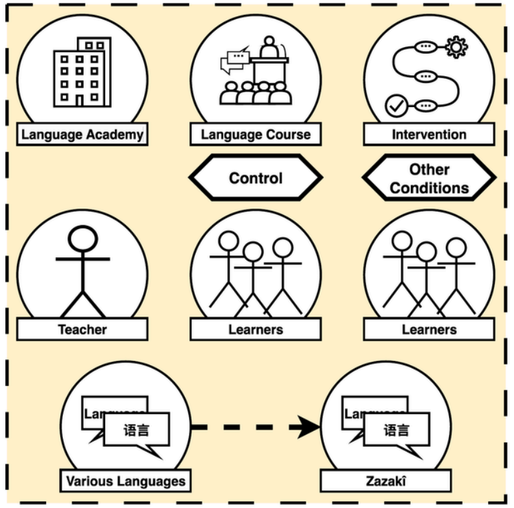
E1
University module



Institutional pressure ●●●
Financial investment ●○○
Voluntariness ●○○

High academic pressure

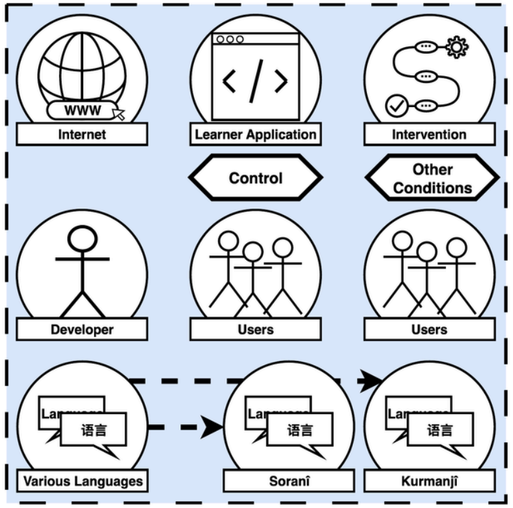
E2
Paid language course



Institutional pressure ●●○
Financial investment ●●●
Voluntariness ●●○

High financial investment

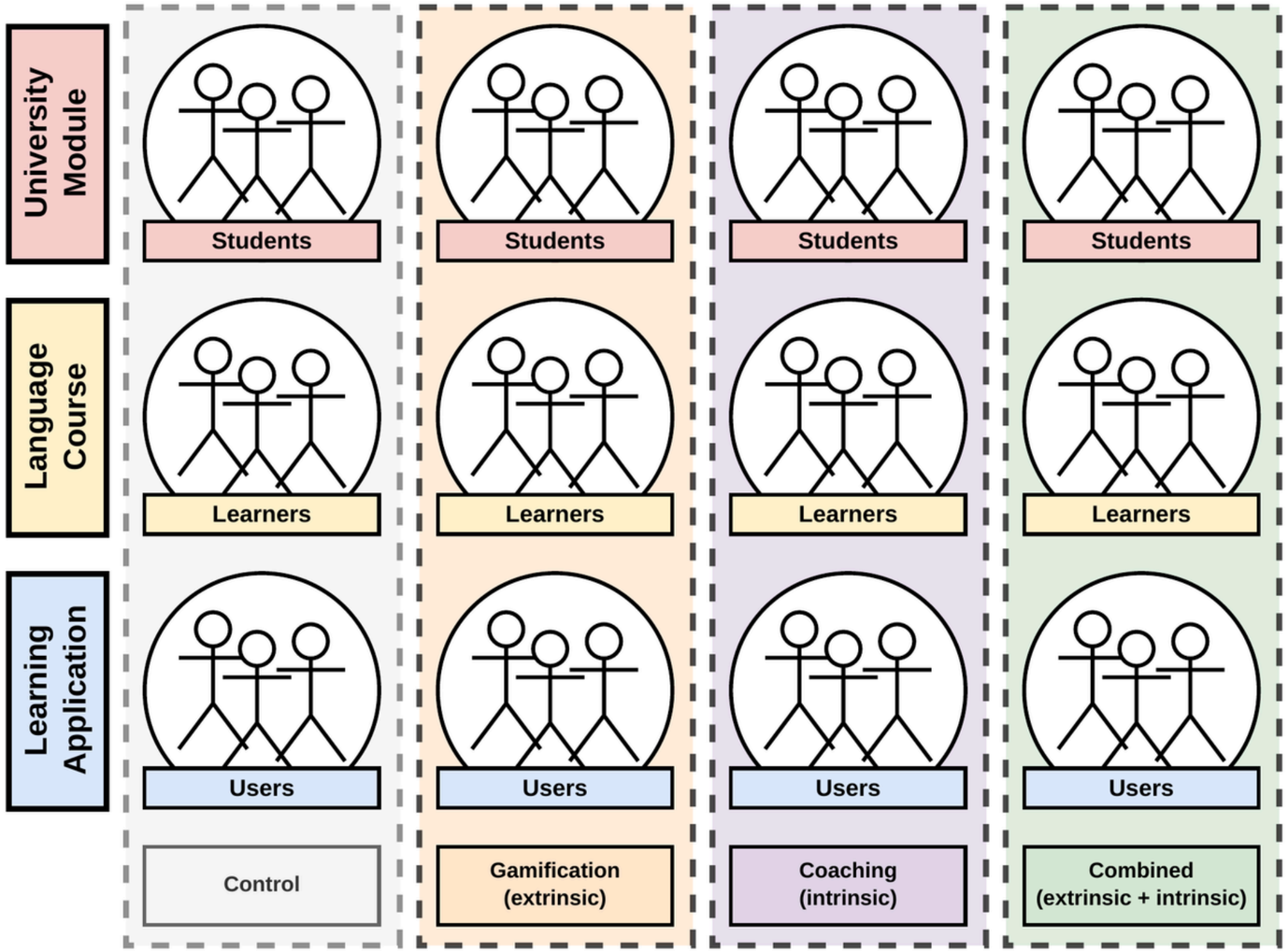
E3
Learning application



Institutional pressure ●○○
Financial investment ○○○
Voluntariness ●●●

Free; can be abandoned any time

Design: 3 environments × 4 conditions



Control
Original activities, digitized but unmodified

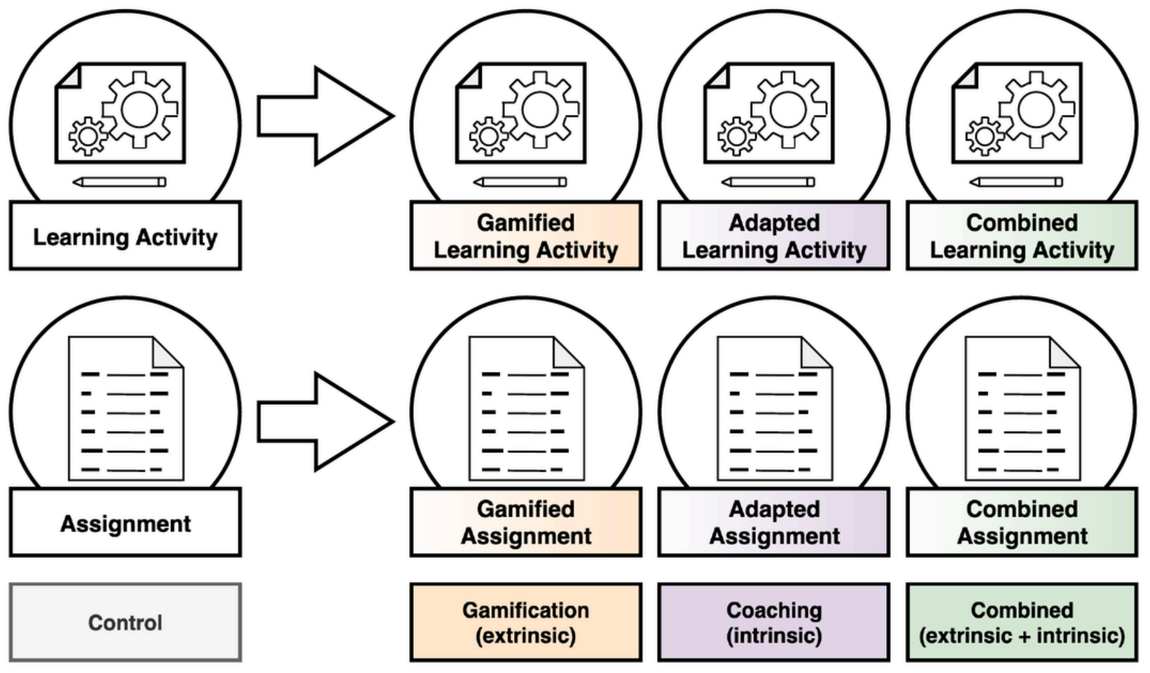
Gamification (extrinsic)
Gameful redesign of the same activities

Coaching (intrinsic)
Agent that prompts changes in learning behavior

Combined
Coaching + the same gamification elements

Within-environment comparison
Content held constant. Environments are not levels of one factor; cross-environment contrast is qualitative.

Gamification shared across environments



Five core learner activities

Vocabulary | Grammar | Alphabet | Listening | Dialogue

Core Gamification Package

Objectives

Missions, side-quests, milestones

Points

XP / points for correct answers or tasks

Progression

Progress bars, steps, maps

Acknowledgement

Badges, achievement markers, trophies

Objectives: personal; others: performance (Toda et al., 2019)

Coaching: a learning companion agent



Timely analytics, personalized guidance and interactive prompts at each stage of the behavior change cycle

Learning analytics per stage

One primary analytic, then learners choose more: descriptive · diagnostic · predictive · prescriptive

On-demand tutor

Help during practice, focused on the long-term routine

Personal companion

Customizable avatar, visual, and style

Background: SRL (Zimmerman, 2000); Stages of Change (Prochaska et al., 1992)

How we will measure the effects

All groups are measured with the same instruments, before and after the intervention period

Learning outcomes

Language proficiency test at start and end, plus shorter tests between condition intervals

What learners say

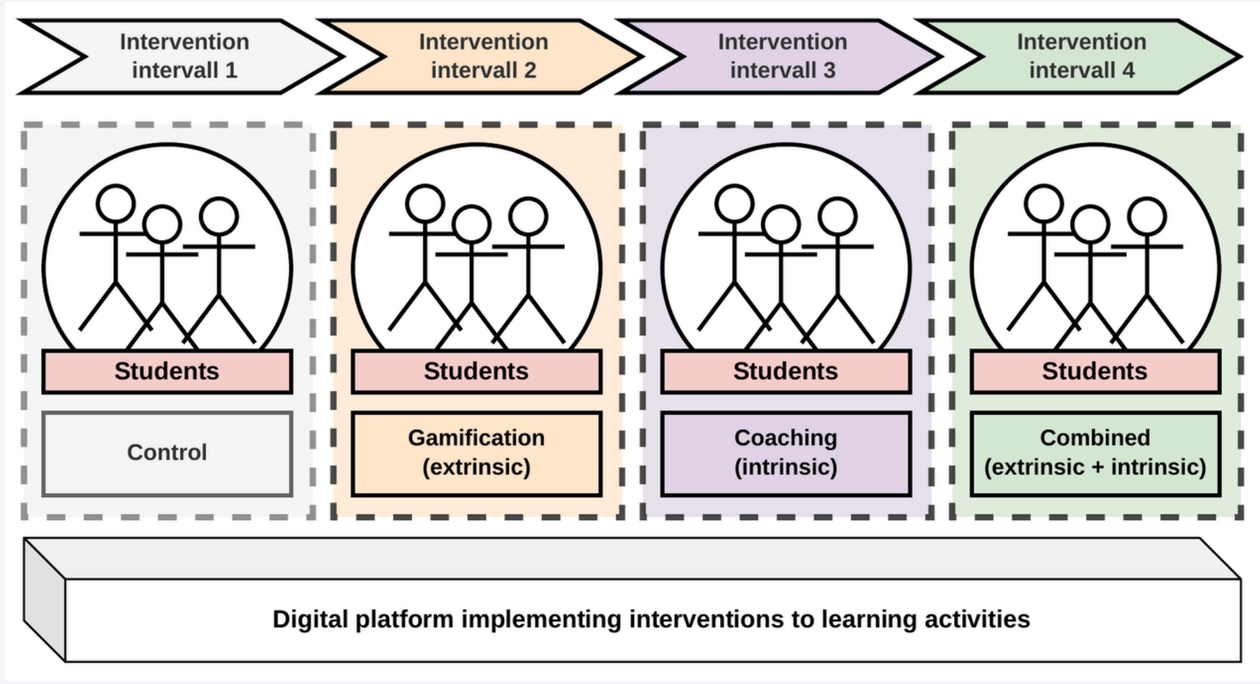
- Self-regulation questionnaire (initial & final)
- Survey on perceived gameful design

What learners do

- Session count
- Time on task
- Feature interactions

Telemetry + self-report → separate reported engagement from actual engagement

Pilot in E1 University: a staged progression



The whole class moves through the four conditions together; measurements at every transition. Chosen because the first cohort is small.

Current status

Now

Prototype close to deployment; digitizing the lecturer's assignments

Until end of 2026

E1 pilot runs in four intervention intervals

Next

E3 (web app): from design to implementation

Then

E2 (language course), building on its quiz app

Limitations



No dedicated funding

We rely on voluntary participation



Small samples

Self-selected groups; attrition may affect comparability



Uneven exposure

In E3 Learning application, users start and stop at any time, so actual exposure to the intervention varies

Further risks

Fixed pilot order

no causal claims per stage

Self-report bias

esp. E1 University, where the lecturer grades

Novelty effects

few measurement points

Transfer

across varieties & instruments



Thank you! 谢谢! سوپاس! Danke! Spas!

A 3 × 4 design comparing gamification and coaching across three Kurdish learning environments, to understand how gameful design and behavioral guidance interact with learner context.

Questions & feedback welcome

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Appendix

Gamification Taxonomy from (Toda et al., 2019)

